THE CURIOSITY SERIES · ARTICLE X

BELIEVE IT OR NOT

THE JOURNEY OF WATER

From molecules and music to copper vessels, mantras, memory and the meanings we give to water.

Water is at once chemistry and culture: it carries dissolved minerals, shapes landscapes, sustains the body, and occupies a sacred place in human life. Across traditions, people have asked whether words, prayer, music or intention can change water in ways we do not yet understand.

This article does not begin by accepting the claim — or dismissing the people who find meaning in it. It begins by separating what can be measured, what can be experienced, and what still needs convincing evidence.

THREE QUESTIONS HIDING INSIDE ONE QUESTION

"Can water respond to intention?" is not one question. It is three, and each one is answered by a different kind of evidence.

1 PHYSICAL WATER

What can be measured?

Molecular structure · temperature · dissolved substances · sound-driven motion

ESTABLISHED SCIENCE

2 THE HUMAN RESPONSE

What can a ritual change in us?

Breathing · attention · expectation · stress · meaning · behaviour

RESEARCHABLE, CONTEXT-DEPENDENT

3 THE MEMORY CLAIM

What remains unproven?

A lasting imprint of words or intention that survives after ordinary physical causes are controlled

NEEDS A MECHANISM AND REPLICATION

> **Sound moving a liquid is not a liquid understanding language. Prayer changing a person is not water storing the prayer.**

The central question breaks into three separate investigations: water's physical behaviour, the effects of ritual on people, and the unproven claim of a lasting intentional imprint.

> THE QUESTION THAT GUIDES THIS JOURNEY
>
> When we say water "responds" to intention, are we describing a physical change in the water, a real change in the person, a cultural or spiritual meaning — or several different things at once?

PART I START WITH WHAT SCIENCE ALREADY KNOWS

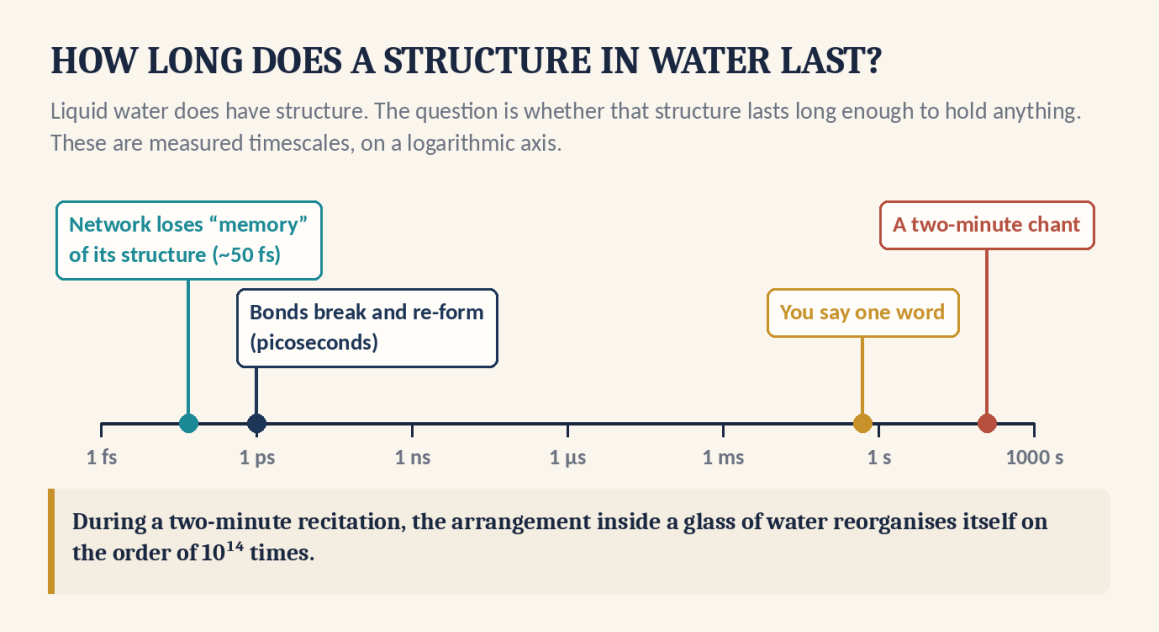
1. A familiar molecule, an extraordinary system

Water is not chemically simple in the sense of being uninteresting. Its molecules form a constantly changing network of hydrogen bonds. Researchers can probe these changes using spectroscopy, scattering experiments and molecular simulations. In one Nature study, ultrafast electron-scattering measurements observed a hydrogen-bond response over femtoseconds and thermalisation on roughly a picosecond timescale [1].

It is worth pausing on what a picosecond is, because the number does the argument's real work. A picosecond is a millionth of a millionth of a second. In the time it takes to say a single word, the hydrogen-bond arrangement inside a glass of water

has reorganised itself something like a million million times. One Nature study of these dynamics found that the network loses its "memory" of its own structural correlations within about fifty femtoseconds [8] — the physicists mean something narrow and technical by the word, but it is exactly the right one. Water genuinely has structure. What it does not obviously have is a structure that stays still.

That is genuine complexity. But complexity is not a blank cheque: **it does not automatically show that water understands language, records emotion, or preserves the meaning of a prayer.**



Measured timescales for structure in liquid water, set against the timescale of human speech.

PART II THE QUESTION MADE FAMOUS BY CRYSTAL PHOTOGRAPHS

2. When water appears to respond to intention

The documentary Every Living Thing, Masaru Emoto's crystal photographs, and Yasuyuki Nemoto's discussions of hado invite a striking possibility: that water exposed to words, music, prayer or intention might form different patterns. In the popular presentation, positive words or music are associated with beautiful, symmetrical crystals; negative words or sounds with less orderly forms.

There is more to report than a simple "debunked" or "proven." A 2006 double-blind pilot study by Radin, Hayssen, Emoto and Kizu reported higher aesthetic ratings for crystal images associated with distant positive intention [2]. A 2008 triple-blind replication by Radin and colleagues also reported a difference in ratings [3]. These are published, positive findings and should be acknowledged.

It is worth stating what those studies actually did, because the design is more interesting than either camp usually admits. In the 2006 pilot, roughly two thousand people in Tokyo directed positive intention toward water held in an electromagnetically shielded room in California. Control samples were kept elsewhere and the senders were not told about them. One hundred independent judges then rated the resulting crystal photographs for aesthetic appeal without knowing which set was which, and the treated images scored higher (reported at $p = 0.001$, one-tailed). That is a real protocol with real blinding, and dismissing it unread is not scepticism — it is just a different kind of incuriosity.



Author-supplied visual panel: the opening of the journey, from water's physical importance to crystal claims and sacred-water traditions.

EVIDENCE IS ABOUT THE MEASURED OUTCOME

3. What a crystal photograph can — and cannot — tell us

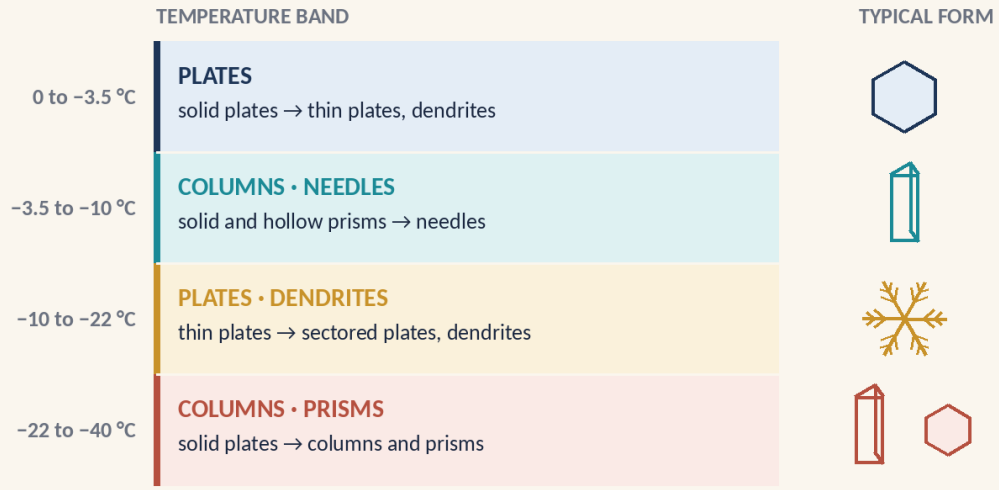
The reported studies assessed the aesthetic appeal of selected crystal photographs. That is not the same as directly measuring a persistent molecular imprint, and it is not proof that water understands the semantic meaning of a word. Crystal formation is sensitive to physical conditions; image selection and subjective ratings also need careful controls.

The variable nobody photographs

"Sensitive to physical conditions" understates it. Ice-crystal morphology is one of the most thoroughly mapped phenomena in atmospheric physics, and it is governed by two variables: temperature and supersaturation. Temperature decides whether a crystal grows as a flat plate or a hollow column — and as the temperature falls from freezing to −40 °C, the habit flips between them three times. Supersaturation decides how branched and ornate the result becomes. A fraction of a degree, or a slight difference in how much moisture is available, moves a sample from one regime to another.

WHAT ACTUALLY DECIDES THE SHAPE OF AN ICE CRYSTAL

Before asking whether a word changed a crystal, it is worth knowing what is already known to change one. Two physical variables — temperature and supersaturation — control the form.



WITHIN EACH BAND · low supersaturation → simple, solid forms | high supersaturation → thin, branched, ornate forms

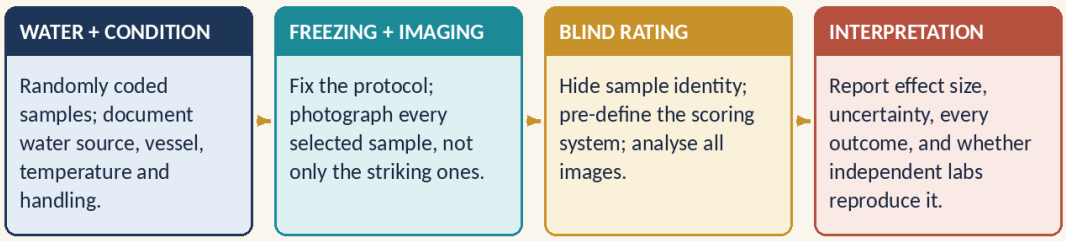
A fraction of a degree moves a sample across a band — which is why controlling temperature is not a technicality here, it is the experiment.

Snow-crystal morphology as a function of temperature and supersaturation — the classical Nakaya diagram, redrawn. Established physics, measured long before the intention experiments [9].

Two consequences follow, and neither is a dismissal. First, two dishes of water that differ by a fraction of a degree will reliably produce different-looking crystals with no intention involved at all — which means that controlling temperature to that precision is not a technicality, it is the whole experiment. Second, fewer than one in a thousand natural snow crystals shows the ideal six-fold symmetry of the famous photographs. Any protocol in which a human being looks at a dish and selects the crystal worth photographing has already made a measurement decision. The 2006 study deserves credit here: its photographer did not know which samples were which, which closes the most obvious door. What blinding the photographer cannot close is a physical difference between the two sets themselves — treated samples and controls were kept in different places, and anything that differs between them, from handling to dissolved impurities to the exact conditions under which each dish froze, is a candidate cause to rule out before intention is.

FROM A BEAUTIFUL PHOTOGRAPH TO A SCIENTIFIC CLAIM

A photograph can raise a question. Only the method decides how much of an answer it gives.



The published studies measured the appeal of a photograph as judged by people — not a physical property of the water.

A rigorous chain from coded samples to independent replication. Each step reduces a different source of bias.

The scientific status of the larger claim remains unsettled. The results justify asking sharper questions; they do not establish a reliable mechanism or a broad scientific consensus. Stronger evidence would require pre-registered protocols, objective outcomes, full reporting of samples, and independent laboratories reproducing the result.

> **THE DISTINCTION THAT MATTERS**
>
> A positive experimental result can be evidence for a specific effect under a specific method. It does not automatically prove the interpretation attached to that effect.

A question worth keeping open

Could a reproducible physical difference be found when intention is tested under strict controls? That is a testable question. "Water remembers how we feel" is a much larger claim, and would need evidence that connects the proposed cause, a measurable water property, and a dependable result.

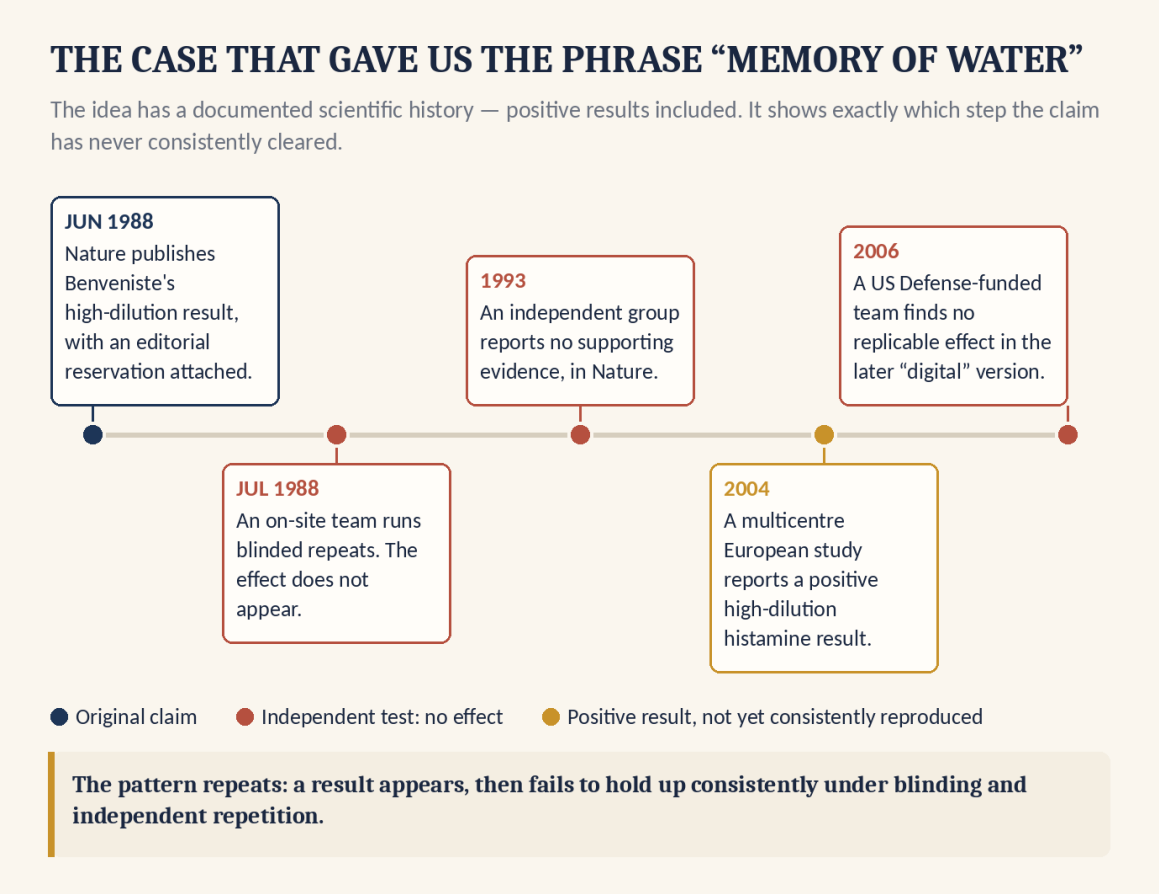
PART III THE CASE THAT NAMED THE IDEA

4. Where the phrase "memory of water" actually comes from

The phrase did not begin with crystal photography. It begins in June 1988, when Nature published a paper from Jacques Benveniste's laboratory reporting that a solution diluted past the point where any original molecule could remain still triggered a biological response [10]. The journal's editor, John Maddox, did something almost unprecedented: he printed the paper with an editorial reservation attached, stating that there was no physical basis for the effect and that Nature had arranged for independent investigators to observe repetitions of the experiment.

They went. During their visit the team watched the assay run, first as the laboratory normally ran it, then under sample codes nobody in the room could read — the code sheet was sealed in foil and taped to the ceiling. Most of the unblinded runs looked favourable. The blinded ones did not. Nature published the team's account the following month [11]. Benveniste never accepted the finding. An independent group reported no supporting evidence in Nature in 1993 [12], and when the work later moved into a "digital" form — the effect supposedly recorded and transmitted electronically — a US Defense–funded team tested that version too and found no replicable effect [13].

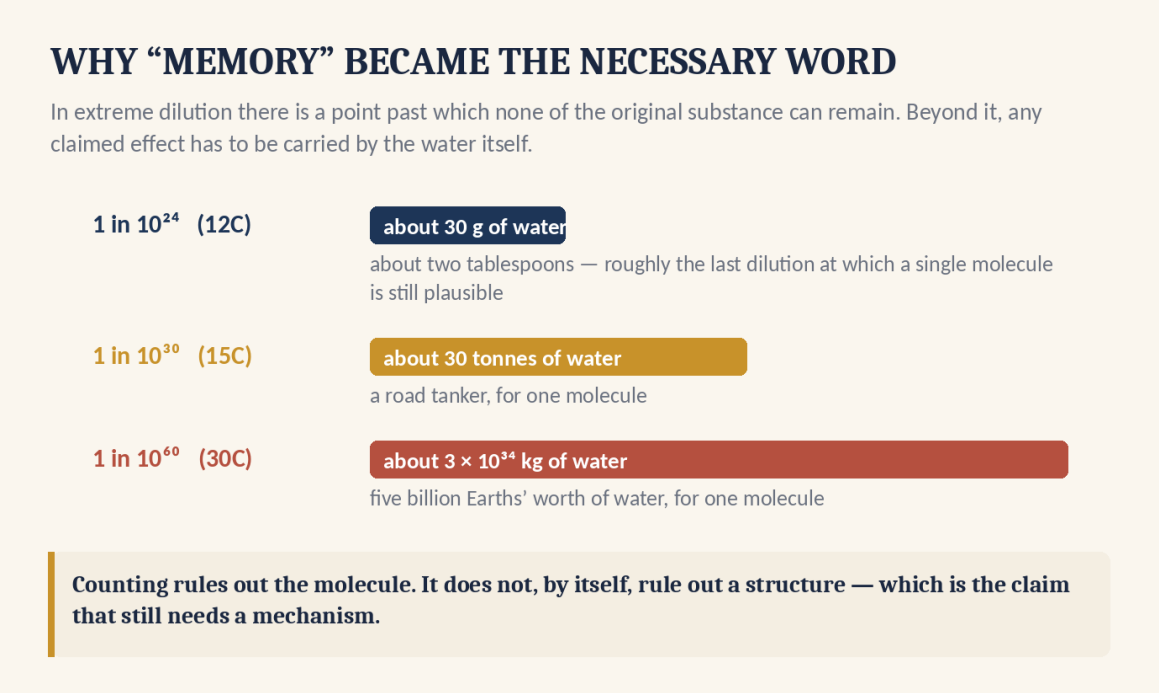
Fairness requires one more entry, because it is the part usually left out. In 2004 a multicentre European study reported that high dilutions of histamine did modulate basophil activation [14] — a positive result in the same family as Benveniste's, from several laboratories. It has not settled the question either. A later replication team noted that the dilution levels reported as active and inactive differed in most studies [15]. An effect that turns up at different dilutions in different hands may still be real — but it is also exactly what noise looks like, and only a protocol that fixes the active dilution in advance can tell the two apart.



The documented history of the water-memory claim, positive results included. The pattern is not failure to publish; it is failure to hold up under blinding and independent repetition.

Why a memory had to be proposed at all

The reason Benveniste needed the word memory is arithmetic, and it is worth doing once. A mole of water contains about 6.022×10^{23} molecules. Each serial hundredfold dilution removes two zeros from the concentration. Somewhere around the twelfth such step, the expected number of original molecules in a glass drops below one. Past that point the substance is not dilute — it is absent. At the thirtieth step, you would need roughly 3×10^{34} kilograms of water, about five billion Earths' worth, to expect a single surviving molecule.



The counting argument, drawn to scale. It is the reason the carrier of the effect had to become the water itself.

This is simple counting, and it cuts in a specific direction. It rules out the molecule. It does not, on its own, rule out a structure — which is exactly the hypothesis that was proposed, and exactly the hypothesis that still has no reproducible mechanism. Martin Chaplin's overview of the debate is the fair place to read both sides of it [7]. Keep this distinction, because it returns later in this article in a gentler form: knowing that something cannot be carried by the obvious mechanism is not the same as knowing it cannot happen.

PART IV MEANING BEFORE MECHANISM

5. The Indian tradition: water as a sacred medium

In Hindu traditions, water is not only a resource. The kamandalu carried by sages and ascetics, the kalasha used in worship, and acts such as abhisheka and ritual sprinkling give water roles in purification, blessing, auspicious beginnings and devotion. Stories of a sage's tapas, sankalpa and spoken words often connect intention with sacred action.

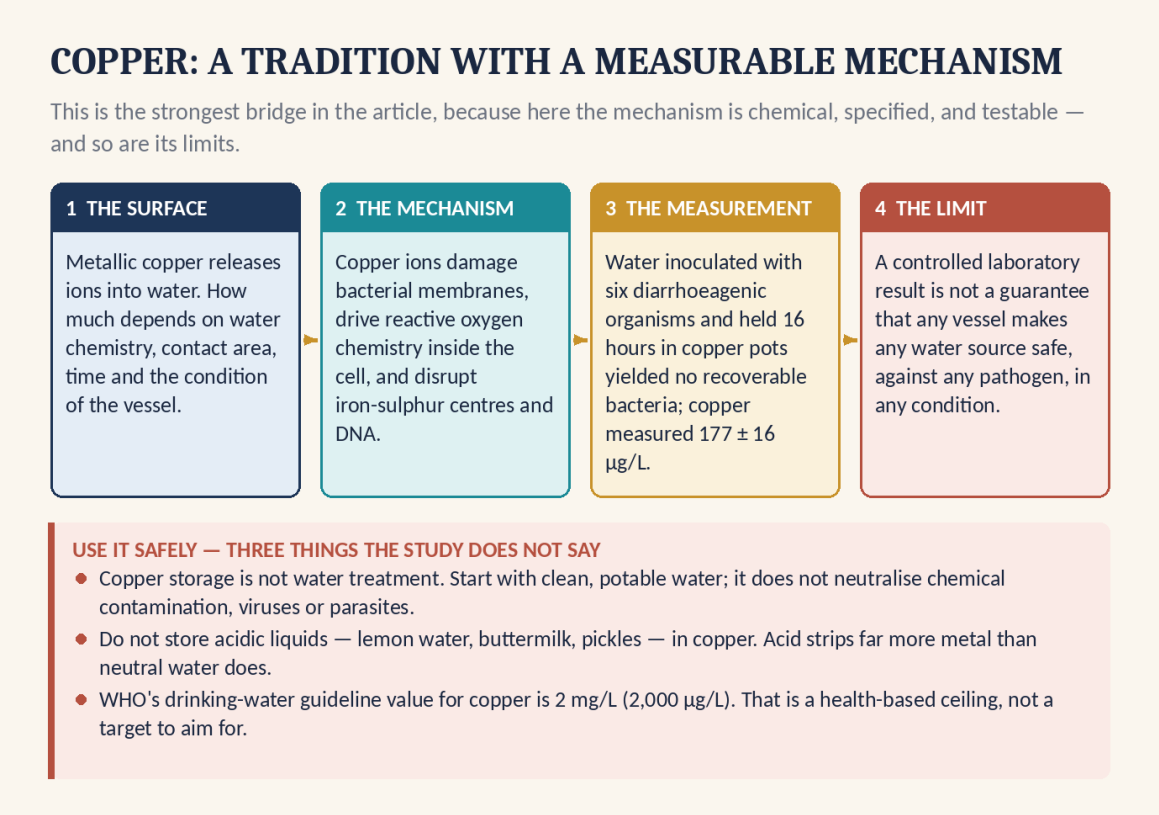
These traditions can be understood in their religious, philosophical and cultural dimensions without pretending that a story is a laboratory demonstration. Their meaning is real as human tradition; a claim that water physically stores a blessing or curse is a separate claim about mechanism.

A MECHANISM RESEARCHERS CAN TEST

6. Copper: one place where tradition meets measurable chemistry

Copper is the clearest example in this journey of a traditional practice meeting a measurable physical property. Copper surfaces can release ions, and copper can inhibit certain microorganisms under suitable conditions. A 2012 laboratory study tested water inoculated with six types of diarrhoeagenic bacteria. After 16 hours in copper pots, the bacteria were not recovered using the reported culture methods; measured copper in the stored water was 177 ± 16 micrograms per litre [4].

What makes this different from every other claim in this article is that the mechanism is named, not merely asserted. Copper ions released into solution attack bacteria along several routes at once: they destabilise the cell membrane, drive reactive oxygen chemistry inside the cell, and disrupt iron–sulphur clusters that the cell needs to run its metabolism [16]. This is why the tradition and the chemistry can sit in the same sentence without embarrassment. Vessel, contact time and water chemistry are all variables you can set, measure and report — and the people who stored water in copper for centuries were, without the vocabulary for it, exploiting a real effect.



Mechanism, measurement and limits: laboratory antimicrobial activity is a narrower claim than universal water purification.

The limit matters as much as the finding. That experiment does not show that every copper vessel makes water from every source safe, or that copper removes every class of pathogen. WHO's copper guideline for drinking water is 2 mg/L (2,000 µg/L), a health-based guideline value — not a target for a vessel and not permission to ignore source-water safety [5]. Use clean, potable water; do not treat copper storage as a substitute for appropriate water treatment.

A claim that looks like copper but is not yet copper

It is worth holding the copper example beside a more recent one, because the difference between them is the whole method of this article. In early 2025, during the Maha Kumbh, widely circulated reports described a finding that the Ganga removes harmful bacteria some fifty times faster than any other freshwater river, attributed to roughly 1,100 types of bacteriophage — viruses that destroy bacteria [17]. The underlying idea is genuinely plausible: bacteriophages are real, they are abundant in river water, and self-purification is a studied phenomenon.

But notice what is missing. None of the circulated reports names a journal, a method, a comparison river or a sample count that others could check. The finding was quickly repeated in political messaging as proof that the river was "completely germ-free," while a Central Pollution Control Board report from the same weeks, cited by the National Green Tribunal, flagged faecal coliform levels at Prayagraj that sit badly with that phrase — a reading state authorities disputed. The copper claim and the phage claim are the same shape: a traditional belief with a candidate mechanism. One has a published protocol, a measured number and a stated limit. The other, so far, has a press report. That is not a reason to decide the phage claim is false. It is the reason it cannot yet be called established.

PART V SOUND, AND THE PERSON MAKING IT

ONE WORD, THREE DIFFERENT MEANINGS

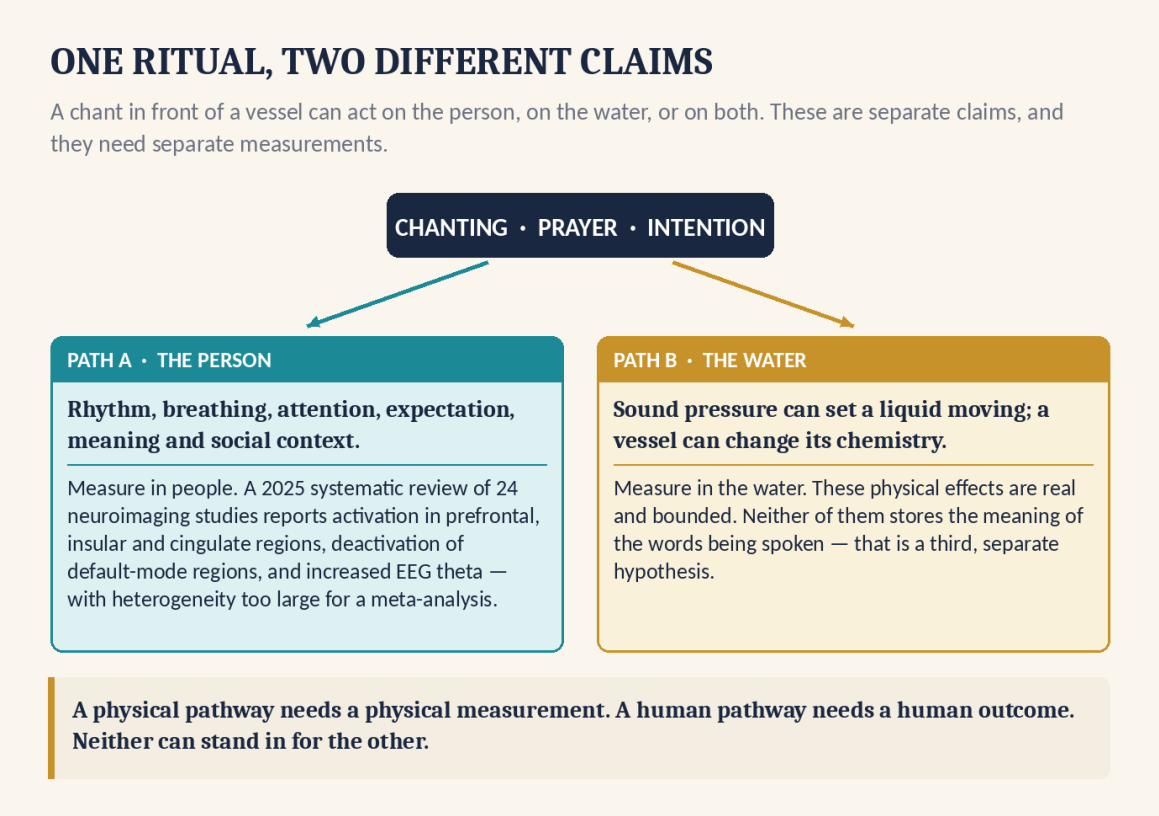
7. Mantras, sound waves and vibration

A chanted mantra is also a physical sound: pressure variations travelling through air. Sound can make nearby objects and liquids vibrate, but the effect depends on frequency, amplitude, distance, vessel shape and the way energy couples into the liquid. Laboratory ultrasound techniques can create effects that ordinary chanting does not automatically reproduce.

The familiar demonstrations in which sand or water organises into geometric patterns on a vibrating plate are real physics, and they are often shown as evidence for something they do not show. Those patterns are standing waves: at a given

frequency, some lines on the surface stay still while the rest moves, and sand gathers along the still lines; on a liquid the same physics draws standing ripples. Change the frequency and the pattern changes. What the demonstration reveals is the geometry of the plate and the frequency applied to it — not the meaning of whatever was said. A signal generator set to the same frequency as a sung note produces the same pattern.

This is why "vibration" needs precision. **In physics it names measurable motion. In spiritual language it may express presence, devotion or inner state. Both uses can be meaningful, but they are not interchangeable.** There is no recognised scientific basis for the online description of the Gayatri Mantra as the "longest recorded frequency in the world"; a chant's frequency depends on how it is performed, and no standard ranking establishes that claim.



The same ritual can have a researchable pathway through human attention and physiology while a lasting change in the water remains a separate hypothesis.

8. What does chanting change in the person?

A 2025 systematic review synthesised 24 neuroimaging studies of chanting, mantra and repetitive prayer. It reported activity in brain regions associated with attention and emotional regulation, while also noting substantial differences among studies that prevented a formal meta-analysis [6]. This is promising, not a final verdict. The evidence concerns people practising the chant — not proof that the water beside them receives or stores its meaning.

The detail is worth having. Across those studies, chanting was associated with activation in prefrontal, insular and cingulate regions — the circuitry of attention and emotional control — and with reduced activity in default-mode regions such as the posterior cingulate cortex, which the authors read as a quieting of self-referential thought. EEG work reported increased theta activity, consistent with a relaxed, inwardly focused state. That is a substantial finding about what the practice does to the practitioner, and it deserves to be stated as plainly as the sceptical point that follows it.



Author-supplied visual panel: copper vessels, sound and the devotional use of water.

RESPECT THE PRACTICE; TEST THE PHYSICAL CLAIM

9. Hanuman Chalisa, sacred water and healing

Imagine a person placing a copper vessel before Hanuman Ji, reciting the Hanuman Chalisa, and later drinking the water as prasad. For the devotee, the act may express courage, protection, gratitude, surrender and hope. It can organise attention and provide a moment of calm, meaning and connection.

If that person feels better afterwards, several explanations may coexist: a condition may be improving naturally; prayer and repetition may change stress, attention or expectation; behaviour or social support may have shifted; or the timing may be coincidental. Personal experience matters, but one experience cannot determine which explanation caused the improvement.

It is easy to hear "expectation" as a polite word for nothing happened. It is not. Expectation is itself a physical mechanism with a measurable signature. In a classic 1978 experiment, placebo pain relief was reversed by naloxone, a drug that blocks the brain's own opioids — belief acting through real chemistry [18]. In a small 2010 trial, patients with irritable bowel syndrome who were told outright that they were taking sugar pills still reported more relief than patients given nothing [19]. So when this article suggests that a ritual may be acting through the person rather than through the water, that is not a smaller claim. It is a different claim, about a different system, requiring a different measurement.

A claim that a ritual-charged water independently treats an illness requires clinical evidence designed for that illness and outcome. Spiritual practice can complement care, but it should not replace diagnosis, prescribed treatment or safe hydration.

> **A RESPECTFUL BOUNDARY**
>
> Faith can remain faith. Science can study measurable effects. The two do not have to be collapsed into one claim for a devotional practice to be meaningful.

The Hanuman Chalisa and the vessel

The act itself is observable: a text is recited, a vessel is present, water is later consumed, and the person interprets the moment through tradition. What is not automatically observable is a new physical property in the water. To establish that, we would have to measure the water independently and show that the change is reproducible.

PART VI FIRST DEFINE WHAT "MEMORY" MEANS

10. Can water remember?

Water certainly carries physical traces of its history. Dissolved minerals, salts, contaminants, isotopes, gases, temperature and contact with a vessel can alter what a sample contains or how it behaves. Those are material records with mechanisms that can be measured.

The stronger claim is different: that ordinary water retains a lasting imprint of a word, emotion or prayer after ordinary physical causes have been controlled. That claim has no established, reproducible mechanism in mainstream chemistry. The hydrogen-bond network of liquid water continually rearranges on femtosecond-to-picosecond timescales [1]. This rapid molecular dynamics does not mean water can retain no physical history; it means that a long-lived "emotional memory" needs a clearly specified mechanism and evidence of its own.

There is a precise way to say this. Water has a memory of its composition — what is dissolved in it, what it has touched, what isotopes it carries. It is not known to have a memory of its information: the meaning of a word spoken over it. The two are different kinds of record, and collapsing them into one word is where most of the confusion in this whole subject begins.



Author-supplied visual panel: water memory, belief and healing, and a framework for connecting different kinds of evidence.

The evidence map

ESTABLISHED. Water has dynamic molecular structure; sound can transfer energy; copper can inhibit some microbes in controlled settings.

PLAUSIBLE / CONTEXTUAL. Chanting and ritual can affect attention, breathing, expectation, stress and the personal meaning of an experience.

NOT ESTABLISHED. That water stores the semantic meaning of words or intention, or that prayer-charged water independently cures disease.

THE EVIDENCE MAP

Every claim in this article, sorted by the strength of evidence behind it rather than by how appealing it is.

ESTABLISHED Supported by measurement and replication	PLAUSIBLE · CONTEXTUAL Real effects, active research, strong dependence on conditions	NOT ESTABLISHED No reproducible mechanism; would need independent replication
• Liquid water has a dynamic hydrogen-bond structure that rearranges on a picosecond scale. • Sound transfers energy into liquids and can set them moving. • Copper releases ions that inhibit certain microbes under controlled conditions.	• Chanting and ritual can change attention, breathing, expectation and felt stress. • Expectation itself has measurable physiological effects. • Vessel, source and storage genuinely change what is in a glass of water.	• That water stores the semantic meaning of words or intention. • That prayer-charged water independently treats disease. • That a specific chant has a uniquely privileged physical frequency.

Nothing in the third column is called false. It is called unsupported — a statement about the evidence, not about the people who hold it.

Every claim in this article, sorted by the evidence behind it rather than by its appeal.

PART VII CURIOSITY DESERVES A STRONG EXPERIMENT

11. If we wanted to test it fairly

A useful experiment would not ask whether "spirituality works" in general. It would define one claim: does a specified sound exposure change a measurable water property? Does coded exposure to intention affect a pre-defined outcome? Does any effect survive blind assessment and independent replication? Those are separate experiments, not one combined test.

WHAT A FAIR TEST WOULD ACTUALLY LOOK LIKE

Do not test "spirituality" as one vague bundle. Test one claim at a time, and decide in advance what would count as a negative result.

1 **DEFINE**	Name one claim. Is it about sound, intention, crystal appearance, chemistry, or health? Each needs a different experiment.
2 **CONTROL**	Use water from a single batch, identical vessels, matched temperature, light, handling and contact time.
3 **RANDOMIZE**	Code the samples. Keep the handler and the analyst blind to which is which.
4 **MEASURE**	Choose an objective outcome in advance. Image every sample, not the striking ones. Publish all results.
5 **PRE-REGISTER**	State the hypothesis, sample size and analysis before collecting data, so the test can fail.
6 **REPLICATE**	An independent laboratory repeats the protocol before any broad conclusion is drawn.

If no imaginable outcome would change the answer, the experiment is a ceremony — a well-meant one, but not a test.

A credible protocol makes the result interpretable whether it is positive, negative or inconclusive.

For an intention experiment, researchers should use water from the same batch, randomise coded samples, standardise the environment, keep handlers and analysts blind where possible, pre-register the outcome, publish all samples and analysis code, and invite independent laboratories to repeat the work. If crystal aesthetics are scored, that should be a secondary, clearly defined measure — not a substitute for a direct physical measurement.

A THREAD FROM THE LAST ARTICLE

Article IX ended on the same test, arriving from the opposite direction: a claim earns the word scientific when you can state, before running the experiment, what result would make you abandon it. Murray's law and the ¾-power scaling in that article passed it: each predicts a number that could have come out wrong. The crystal photographs can be made to pass it too — but only by someone willing to specify, in advance, the outcome that would end the idea.

THE POSITION THIS ARTICLE TAKES

The dots connect — but not into a single conclusion

Water is biological necessity, chemical medium, cultural symbol and spiritual offering. Copper gives us a clear example of a traditional material with a measurable antimicrobial mechanism. Chanting gives us a way to study sound and human physiology. Sacred-water traditions show how communities express purification, blessing and devotion. These connections are real — but they do not all answer the same question.

Some findings are established. Some interpretations are meaningful within a tradition. Some hypotheses remain open, and some are not supported by reproducible evidence yet. The honest position is neither automatic belief nor reflexive dismissal. It is to keep each claim at the level the evidence can support.

BELIEVE IT OR NOT

Let every dot remain a dot until the evidence connects it. Curiosity is not the promise that every mystery will confirm our beliefs; it is the willingness to investigate without deciding the answer in advance.

What I am still curious about

- If temperature and supersaturation were controlled to a hundredth of a degree, and every crystal in the dish were photographed rather than the best one, would any difference remain?
- How much of what a devotee experiences after a ritual is the practice, how much is the expectation, and is that distinction even meaningful to the person living it?
- Why does the structural version of the memory hypothesis keep attracting serious people, nearly four decades after the first blinded test?

This article is an exploratory reflection, not medical advice. Sacred water and devotional practices should not replace potable water, medical diagnosis or evidence-based treatment.

READ BEYOND THE ARTICLE

Selected sources & further investigation

[1] Yang, J., Dettori, R., Nunes, J. P. F., et al. (incl. Gaffney, K. J., Martinez, T. J., Wang, X.). Direct observation of ultrafast hydrogen-bond strengthening in liquid water. Nature 596(7873): 531–535 (2021). Read source

[2] Radin, D., Hayssen, G., Emoto, M. & Kizu, T. (2006). Double-blind pilot test of distant intention and water crystal formation; aesthetic ratings were the outcome. Explore (NY) 2(5): 408–411. Roughly 2,000 senders in Tokyo; 100 blinded judges. Read source

[3] Radin, D., Lund, N., Emoto, M. & Kizu, T. (2008). Triple-blind replication reported in the Journal of Scientific Exploration 22(4): 481–493; read methods and limitations alongside the positive result.

[4] Sudha, V. B. P., Ganesan, S., Pazhani, G. P., Ramamurthy, T., Nair, G. B. & Venkatasubramanian, P. (2012). Storing drinking-water in copper pots and recovery of selected diarrhoeagenic bacteria. J Health Popul Nutr 30(1): 17–21. PMID 22524115. Read source

[5] World Health Organization — Copper fact sheet (current guidance). Copper in drinking-water: guideline value and health basis. Guideline value 2 mg/L.

[6] Perry, G., Wolff, L., Powers, N., Thompson, W. & Baumann, O. (published 2025; indexed 2026). Systematic review of neural correlates of chanting; 24 studies and methodological heterogeneity. WIREs Cognitive Science 16(6): e70018. Read source

[7] Chaplin, M. F. (2007). The memory of water: overview of the term and the debate around proposed mechanisms. Homeopathy 96(3): 143–150.

[8] Cowan, M. L., Bruner, B. D., Huse, N., Dwyer, J. R., Chugh, B., Nibbering, E. T. J., Elsaesser, T. & Miller, R. J. D. (2005). Ultrafast memory loss and energy redistribution in the hydrogen bond network of liquid H_2O. Nature 434(7030): 199–202. Read source

[9] Libbrecht, K. G. The physics of snow crystals / Snow Crystals — reviews of the Nakaya morphology diagram. Temperature sets plate-versus-column habit; supersaturation sets morphological complexity.

[10] Davenas, E., Beauvais, F., Amara, J., et al. & Benveniste, J. (1988). Human basophil degranulation triggered by very dilute antiserum against IgE. Nature 333(6176): 816–818. Published with an editorial reservation (Nature 333: 787).

[11] Maddox, J., Randi, J. & Stewart, W. W. (1988). "High-dilution" experiments a delusion. Nature 334(6180): 287–290, 28 July 1988.

[12] Hirst, S. J., Hayes, N. A., Burridge, J., Pearce, F. L. & Foreman, J. C. (1993). Human basophil degranulation is not triggered by very dilute antiserum against human IgE. Nature 366(6455): 525–527.

[13] Jonas, W. B., Ives, J. A., Rollwagen, F., et al. (2006). Can specific biological signals be digitized? FASEB Journal 20(1): 23–28. US Department of Defense–funded independent test; no replicable effect found.

[14] Belon, P., Cumps, J., Ennis, M., Mannaioni, P. F., Roberfroid, M., Sainte-Laudy, J. & Wiegant, F. A. C. (2004). Histamine dilutions modulate basophil activation. Inflammation Research 53(5): 181–188. Multicentre positive result. Read source

[15] Guggisberg, A. G., Baumgartner, S. M., Tschopp, C. M. & Heusser, P. (2005). Replication study concerning the effects of homeopathic dilutions of histamine on human basophil degranulation in vitro. Complementary Therapies in Medicine 13(2): 91–100. Read source

[16] Grass, G., Rensing, C. & Solioz, M. (2011). Metallic copper as an antimicrobial surface. Appl Environ Microbiol 77(5): 1541–1547. Contact killing and ionic copper toxicity. Read source

[17] Press reports of a Ganga bacteriophage self-purification finding (February 2025), attributed to Dr Ajay Sonkar. Included as a worked example of an unverified claim: the circulated reports name no journal, method or comparison data. Read alongside Central Pollution Control Board water-quality reporting for the same period.

[18] Levine, J. D., Gordon, N. C. & Fields, H. L. (1978). The mechanism of placebo analgesia. The Lancet 312(8091): 654–657. Read source

[19] Kaptchuk, T. J., Friedlander, E., Kelley, J. M., et al. (2010). Placebos without deception: a randomized controlled trial in irritable bowel syndrome. PLoS ONE 5(12): e15591. 80 patients; a small trial the authors say needs larger confirmation.

[20] WHO Guidelines for Drinking-water Quality. Copper guideline value table in the WHO guideline material.

[21] Every Living Thing. Documentary listing referenced in the author's original draft.

[22] Yasuyuki Nemoto. Material on "hado" referenced in the author's original draft.

Editorial note. The two crystal papers are included because they report positive results relevant to the hypothesis, not because they settle it. Their methods and conclusions should be judged separately from the broader claim that water stores emotional meaning. Sources [1]–[7] keep the author's original numbering; [8]–[19] were added in this edition to place the question in its documented scientific history and supply the established physics; [20]–[22] are the author's original further-reading entries.